

satellite. It has also announced plans of further deep space exploration using this lunar observatory as the base.

India

One of the main reasons that China has accelerated its space programme is because of the emergence of India as a viable competitor in the space field. Indeed, some experts are calling the competition between these two emerging economies the New Space Race.

India's indigenous efforts towards a space programme have been fairly recent. Of course, it won't be the first Indian to go into space, which is an honour held by Wing Commander Rakesh Sharma of the Indian Air Force, who went aboard a Soyuz spacecraft to the Salyut 7 satellite in 1984.

The country's main space agency, the Indian Space Research Organisation (ISRO), has announced that it plans to send a man into lower-orbit by 2016, putting it a fair distance behind the Chinese programme. However, it also plans on sending a man onto moon by 2020 – an ambitious project, given the time period and its current state. The project did receive a boost, though, when ISRO successfully launched the Chandrayaan lunar probe.

ISRO is currently working on developing a spacecraft based partially on the Mercury spacecraft and partially on the Soyuz, which will carry a two-member crew into orbit. Since the space agency hasn't released any official specifications about the device, information is available only through different news reports and hence is a bit sketchy. Expected to weigh about 4 ton kilograms, it will be launched by the GSLV-MK-III rocket launcher.

The space capsule will have life control and environment control systems, along with emergency escape systems more advanced than the Soyuz spacecraft. An illustration of the spacecraft that ISRO chairman G. Madhavan Nair showed in a presentation was interpreted by space site Astronautix as depicting a main engine and smaller orientation engines arranged in a light package around the base of the capsule, indicating an earth-orbit manoeuvring capability was to be included. The nose of the original version of the OV was free for a docking mechanism, but primary entry was evidently through a side hatch secured by explosive bolts.

ISRO is currently working to set up an institute to train astronauts in Bangalore.

Private programmes

Apart from China and India, other countries like Iran, Malaysia and Ecuador